

AskMyDocs Answer

Query: what is Transformer and RAG

Based on the provided context, here's what I can tell you about Transformers and RAG:

Transformer: The Transformer is a neural network architecture that improves upon traditional sequence-to-sequence (Seq2Seq) models by:

- Enabling parallel processing through self-attention mechanisms (unlike sequential RNNs)
- Capturing long-range dependencies between tokens
- Using positional encodings to preserve sequence order
- Implementing multi-head attention to focus on different aspects of the input simultaneously
- Addressing the vanishing gradient problem through self-attention, residual connections, and layer normalization

Transformers use attention mechanisms that compute similarity scores between query, key, and value vectors to focus on relevant tokens in a sequence. This makes them highly effective for NLP tasks.

RAG (Retrieval-Augmented Generation): Retrieval-Augmented Generation (RAG) is a framework that combines document retrieval with text generation. It works through three specific steps:

1. **Retrieval:** Fetching relevant documents using query embeddings
2. **Ranking:** Sorting documents by relevance
3. **Generation:** Using the retrieved context to generate accurate responses

RAG enhances factual accuracy in tasks like question answering by allowing models to ground their responses in relevant external knowledge rather than just relying on their pre-trained knowledge.

The context shows that RAG is particularly useful for improving factual accuracy in applications where the model needs to access up-to-date information from external sources.